



- 7 The curve  $C$  has equation  $y = \frac{10x^2 - 11x - 18}{10x - 18}$ .

(a) Find the equations of the asymptotes of  $C$ . [3]

This image shows a full page of white paper with ten horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and extend across the entire width of the page. There is no text or other markings on the paper.

(b) Show that  $C$  has no stationary points. [4]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



(c) Sketch  $C$ , stating the coordinates of the points of intersection with the axes.

[3]

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(d) Sketch the curve with equation  $y = \left| \frac{10x^2 - 11x - 18}{10x - 18} \right|$ .

[2]





- (e) Given that  $\left| \frac{10x^2 - 11x - 18}{10x - 18} \right| < 4$  for  $\frac{-29 - \sqrt{4441}}{20} < x < p$  and  $\frac{-29 + \sqrt{4441}}{20} < x < q$ , find the values of  $p$  and  $q$ . [4]

[illegible]